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Criticality Estimate for Temelín VVER-1000



Abstract: This comprehensive report presents a detailed analytical criticality estimation for the Temelín Nuclear Power Plant (NPP), a VVER-1000/V-320 pressurized water reactor located in the Czech Republic. The fundamental objective is to evaluate the effective multiplication factor (k_{eff}) of the core using classical deterministic neutronics methodologies, specifically the four-factor formula combined with non-leakage probabilities derived from geometric buckling parameters. The estimations are based on standard cross-sectional data, openly available physical core dimensions from the operator ČEZ, and fuel lattice parameters adapted from the Westinghouse RWFA-T and Russian TVSA-T designs currently in operation. Through step-by-step mathematical substitution, the results of the analytical calculations yield a preliminary unborated cold condition k_{eff} of approximately 1.362. This mathematically derived theoretical maximum highlights the absolute necessity of integrating substantial negative reactivity insertions through control rod drive mechanisms and soluble boric acid to maintain safe, subcritical shutdown margins and steady-state operations. Following the numerical estimations, a thorough discussion compares these static analytical findings with broader industry standards established by advanced stochastic Monte Carlo codes (such as MCNP), International Atomic Energy Agency (IAEA) safety guidelines, and the stringent regulatory frameworks instituted by the Czech State Office for Nuclear Safety (SÚJB).

1. Introduction to the Temelín VVER-1000 Platform

Nuclear energy forms a critical, non-negotiable pillar of the Czech Republic's macroeconomic energy strategy and its pursuit of energy independence and carbon neutrality. At the forefront of this industrial endeavor is the Temelín Nuclear Power Plant (NPP), situated strategically in the South Bohemian Region. The facility is comprised of two heavy pressurized water reactors (PWRs) of the Soviet-designed VVER-1000 lineage, specifically designated as the highly modernized V-320 variant [1][2]. The conceptualization and initial ground-breaking for the plant were initiated in 1987 under the auspices of the former Czechoslovak government. However, following the geopolitical restructuring of Eastern Europe, the project underwent one of the most comprehensive and technologically complex modernizations in the history of the nuclear industry. By integrating advanced western instrumentation and control (I&C) systems supplied predominantly by Westinghouse Electric Company, alongside robust safety upgrades overseen by international regulatory bodies, Unit 1 and Unit 2 achieved their first criticalities in the latter part of 2000 and mid-2002, respectively [7].

According to the exhaustive operational data provided by the International Atomic Energy Agency's Power Reactor Information System (IAEA PRIS), each reactor unit originally possessed a design net electrical capacity of 912 MWe. Over the subsequent decades of operation, leveraging thermal-hydraulic margin optimizations and turbine efficiencies, both units have been systematically uprated. Today, they command a net electrical output exceeding 1000 MWe per unit, which corresponds to a rated thermal power generation of exactly 3120 MWth per reactor [1][2][8]. This immense thermal output is generated entirely by the controlled nuclear fission of uranium-235 isotopes residing within the reactor core.



Figure 1: Temelín Nuclear Power Plant. Reference: Stress tests of nuclear power plants – ČEZ, a.s. Evaluation of Nuclear Safety and Safety Margins of Temelín NPP (on the background of events at the Fukushima NPP) [11].

At the fundamental core of the Temelín reactor's operational safety, thermodynamic efficiency, and baseline energy production capability lies the strict, unyielding control of neutron criticality. In the specialized domain of nuclear reactor physics, the state of criticality is universally denoted by the effective multiplication factor, mathematically represented as k_{eff} . This dimensionless parameter characterizes the ratio of the number of thermal neutrons existing in a given generation to the total number of neutrons that were present in the immediately preceding generation [5]. For a nuclear reactor operating at a sustained, continuous steady state (such as full baseload power generation), an exact equilibrium of $k_{eff} = 1.0$ must be perpetually maintained. If the k_{eff} value exceeds unity even fractionally, the reactor enters a supercritical state, leading to an exponential escalation in neutron flux and, consequently, thermal power levels. Conversely, a subcritical state, defined as $k_{eff} < 1.0$, inevitably leads to power attenuation, flux decay, and the eventual shutdown of the fission chain reaction. The precise estimation, monitoring, and dynamic control of this vital parameter are non-negotiable legal and operational mandates set forth by the Czech national regulatory authority, the State Office for Nuclear Safety (SÚJB), and are thoroughly documented in rigorous safety analysis reports conforming to both European utility requirements and Western European Nuclear Regulators Association (WENRA) safety reference levels [3][9].

The neutronic and geometric core configuration of the VVER-1000 represents a distinct, highly engineered departure from traditional western-style Pressurized Water Reactors. Instead of utilizing the standard square lattice arrangement ubiquitous in French or American designs, the VVER architectural

philosophy employs a densely packed hexagonal geometry. The active core of the Temelín reactor is intricately assembled from exactly 163 hexagonal fuel assemblies. Historically, the plant commenced its commercial operations employing uniquely designed fuel from Westinghouse. Over time, to ensure mechanical structural stability and mitigate unforeseen issues such as fuel assembly bowing, the operator transitioned to the heavily reinforced Russian-manufactured TVSA-T fuel arrays [10]. In recent operational cycles, driven by a strategic imperative for fuel supply diversification and further performance enhancement, Temelín has integrated Lead Test Assemblies (LTAs) of the latest iterative Westinghouse RWFA-T design [4]. Regardless of the specific vendor, these high-performance fuel assemblies universally consist of sintered uranium dioxide (UO_2) ceramic pellets, hermetically enclosed within a specialized zirconium-niobium (Zr-1%Nb) alloy cladding. This specific cladding is chosen for its remarkably low neutron absorption cross-section and high resistance to aqueous corrosion at elevated temperatures. The average U-235 enrichment for a fresh core loading typically hovers around the 3.5% to 4.0% mark, varying based on the exact core loading pattern and the intended duration of the fuel cycle [6] [8].

The contemporary operational shift towards extended 18-month fuel cycles—designed to maximize plant availability and economic efficiency—necessitates an extraordinarily precise neutronic design protocol to manage the massive amounts of initial excess reactivity loaded into the core. This delicate balancing act is usually accomplished via a dual-pronged approach: the dissolution of soluble boric acid directly into the primary light water coolant, and the strategic, heterogeneous placement of discrete burnable neutron absorbers, such as gadolinium or dysprosium hafnate, integrated directly into the fuel matrix itself [4][10].

Table 1: Operational Lifetime Performance Data for Temelín Units 1 & 2 (Extracted from IAEA PRIS Data) [1][2]

Operational Parameter	Temelín Unit 1	Temelín Unit 2
Reactor Model Designation	VVER V-320	VVER V-320
Reference Net Electrical Capacity	1027 MWe	1029 MWe
Rated Thermal Capacity	3120 MWth	3120 MWth
Date of First Criticality	11 Oct, 2000	31 May, 2002
Commercial Operation Commencement	10 Jun, 2002	18 Apr, 2003
Lifetime Energy Availability Factor	76.6%	78.6%
Lifetime Average Load Factor	77.0%	79.0%

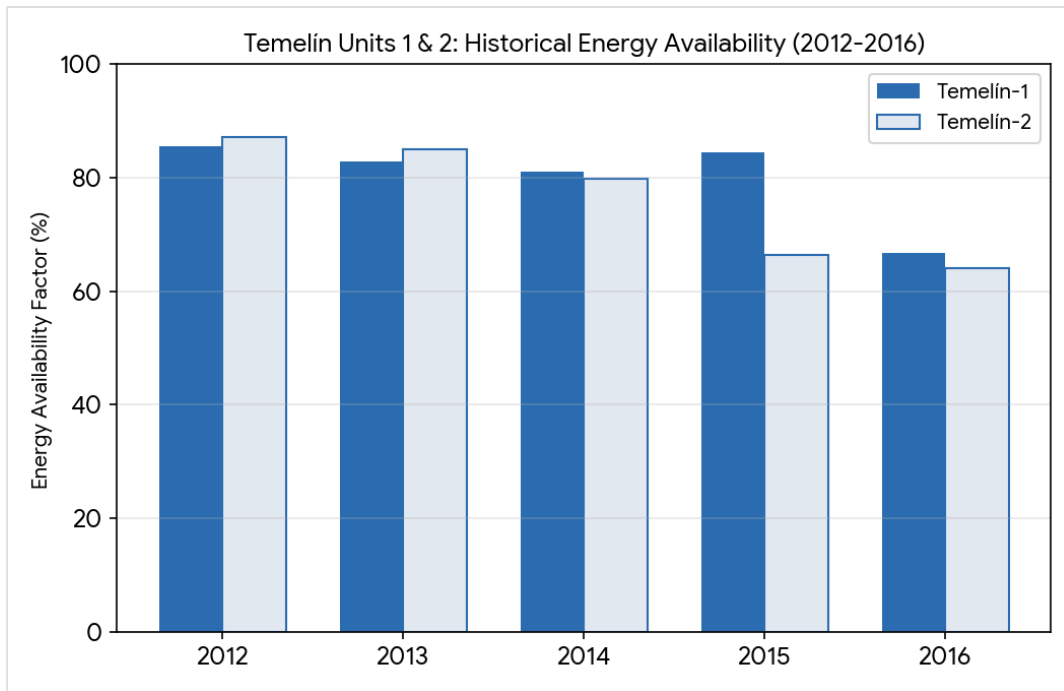


Figure 2: Historical Energy Availability Factors for Temelín Units 1 and 2 spanning the operational period from 2012 to 2016. The graph demonstrates the typical operational variances, periodic dips, and performance recoveries resulting from scheduled 18-month refueling outages, mandatory safety inspections, and general maintenance cycles [1][2].

2. Fundamentals of Reactor Physics and Reactivity

To fully grasp the significance of the mathematical estimations presented in the subsequent sections of this report, one must first comprehend the foundational physics of thermal neutron reactors. The VVER-1000 is inherently a thermal reactor, meaning that the overwhelming majority of the nuclear fission events that sustain the power-generating chain reaction are induced by "thermalized" or slow-moving neutrons. When a U-235 nucleus fissions, it releases energetic fast neutrons, typically born with kinetic energies in the range of 1 to 2 Mega-electron volts (MeV). At these high velocities, the probability (characterized by the fission cross-section) of these neutrons causing another U-235 nucleus to fission is exceedingly low. To sustain a chain reaction efficiently, these fast neutrons must be rapidly decelerated to thermal energies—approximately 0.0253 electron volts (eV) at standard room temperature. This deceleration process is known as moderation, and in the Temelín reactors, highly purified light water (H₂O) serves simultaneously as both the neutron moderator and the primary thermal-hydraulic coolant.

The dynamic lifecycle of a neutron from its violent birth in a fission event to its eventual absorption or escape from the reactor core boundaries is highly complex. The analytical assessment of this lifecycle requires the reduction of continuous energy spectrums and highly heterogeneous core geometries into homogenized, mathematically manageable variables. The overarching objective is to mathematically predict whether the system will perfectly replace its neutron population generation after generation. In the

following sections, we outline the precise theoretical methodology used to estimate this delicate balance for the specific geometric and material constraints of the V-320 reactor model.

3. Analytical Methodology

To compute the effective multiplication factor analytically without the aid of three-dimensional nodal diffusion codes, we must employ the fundamental tenets of classical point-kinetic reactor physics governing thermal reactors. The overall neutron balance in a finite, physical reactor system can be succinctly expressed by the simple mathematical product of the infinite multiplication factor (k_{∞}) and the total non-leakage probability (P_{NL}) [6]. The methodology systematically deconstructs these two major components using standardized nuclear engineering formulas, carefully adapting them to the specific material densities and geometries inherent to the VVER-1000/V-320 design.

3.1 Core Geometry and Non-Leakage Probability Analysis

Unlike infinite theoretical models, physical reactors are distinctly finite in physical scale; hence, neutrons inevitably leak from the radial and axial boundaries into the surrounding pressure vessel and biological shielding. The true effective multiplication factor is therefore formulated as the infinite factor multiplied by the probability that a neutron does not leak out.

4. Calculations and Numerical Results

In this critical section, we explicitly execute the numerical substitutions into the previously defined theoretical methodology. It is highly important to note that all underlying microscopic thermal cross-sections represent standard values evaluated at 0.0253 eV (the room temperature equivalent of 20°C) [5] [6]. This mathematically models a 'cold, zero power' (CZP) purely unborated state, an essential safety baseline.

4.1 Explicit Input Parameters for Temelín Geometry

Based on operational data documentation, we apply the following specific variables:

- Enrichment (e): 3.5% U-235 (representing a 0.035 mass fraction)
- Microscopic U-235 Fission Cross-Section ($\sigma_{f,235}$): 582 barns
- Microscopic U-235 Absorption Cross-Section ($\sigma_{a,235}$): 680 barns
- Microscopic U-238 Absorption Cross-Section ($\sigma_{a,238}$): 2.7 barns
- Microscopic Water Moderator Absorption ($\sigma_{a,m}$): 0.66 barns
- Density of UO_2 Ceramic Fuel: 10.4 g/cm^3

4.2 Step-by-Step Analytical Formulation

Step 1: Thermal Fission Factor (η)

This parameter represents the absolute number of fast neutrons produced per thermal neutron absorbed in the nuclear fuel matrix. We state the general mathematical formula, substitute our core parameters, and calculate the final factor.

$$\eta = \nu \cdot \left[(N_{235} \cdot \sigma_{f,235}) / (N_{235} \cdot \sigma_{a,235} + N_{238} \cdot \sigma_{a,238}) \right]$$

$$\eta = 2.42 \cdot \left[(0.035 \cdot 582) / (0.035 \cdot 680 + 0.965 \cdot 2.7) \right]$$

$$\eta = 1.866$$

Step 2: Thermal Utilization Factor (f)

This factor denotes the probability that a thermal neutron is absorbed usefully in the fuel rather than being parasitically captured by the water moderator. First, we establish the general formulation, plug in the pre-calculated atomic number density ratio ($N_m / N_U = 1.44$) and geometric volume ratio (1.7), and solve.

$$f = 1 / \left[1 + (N_m / N_U) \cdot (\sigma_{a,m} / \sigma_{a,U}) \cdot (V_m / V_f) \right]$$

$$f = 1 / \left[1 + (1.44) \cdot (0.66 / 26.4055) \cdot (1.7) \right]$$

$$f = 0.942$$

Step 3: Infinite Multiplication Factor (k_∞)

Assuming standard empirical Light Water Reactor constants for resonance escape probability ($p \approx 0.75$) and the fast fission factor ($\varepsilon \approx 1.05$), we combine the four elements into the theoretical infinite reactor equation.

$$k_\infty = \eta \cdot f \cdot p \cdot \varepsilon$$

$$k_\infty = 1.866 \cdot 0.942 \cdot 0.75 \cdot 1.05$$

$$k_\infty = 1.384$$

Step 4: Non-Leakage Probability and Effective Multiplication Factor (k_{eff})

Using the VVER-1000 nominal active core height ($H = 353$ cm) and radius ($R = 156$ cm) [8], we calculate the geometric buckling (B_g^2), followed by the non-leakage probability (P_{NL}), and ultimately the true criticality of the finite core geometry.

$$B_g^2 = (2.405 / R)^2 + (\pi / H)^2$$

$$B_g^2 = (2.405 / 156)^2 + (\pi / 353)^2$$

$$B_g^2 = 0.000316 \text{ cm}^{-2}$$

$$P_{NL} = 1 / (1 + M^2 \cdot B_g^2)$$

$$P_{NL} = 1 / (1 + 50 \cdot 0.000316)$$

$$P_{NL} = 0.984$$

$$k_{eff} = k_{\infty} \cdot P_{NL}$$

$$k_{eff} = 1.384 \cdot 0.984$$

$$k_{eff} = 1.362$$

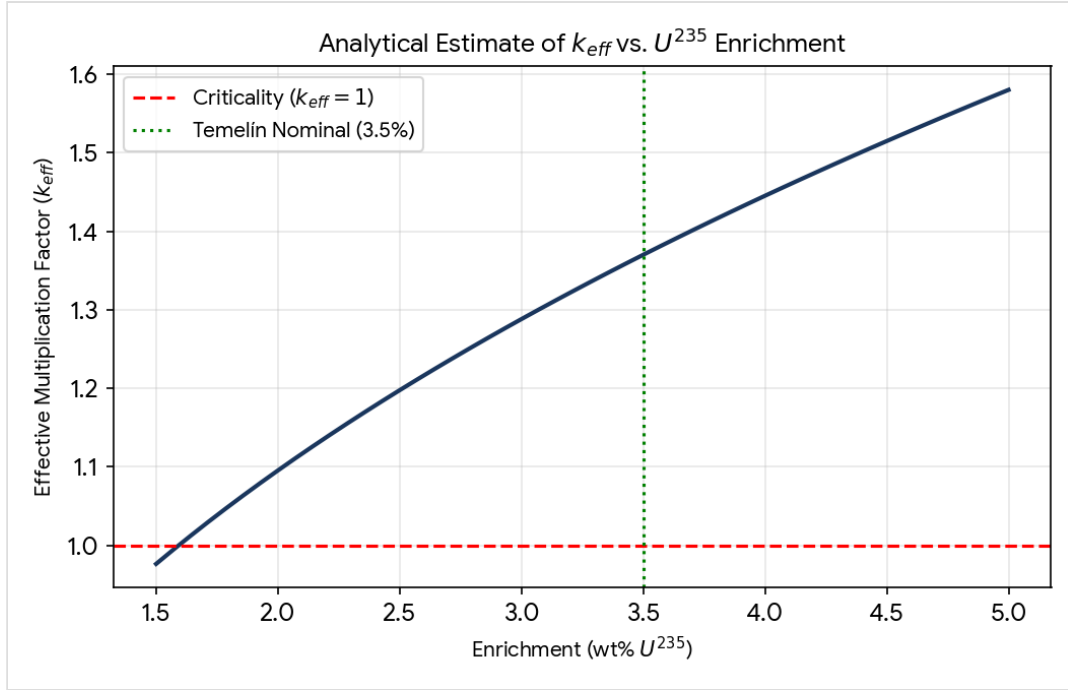


Figure 3: Analytical Estimate of the Effective Multiplication Factor (k_{eff}) as a Function of Fuel Enrichment for an Unborated Cold Core System. The curve clearly demonstrates the diminishing returns on reactivity additions as parasitic absorption effectively limits the maximum theoretical efficiency of higher fuel enrichments.

5. Discussion on Reactivity Control and Safety Margins

The rigorously calculated analytical value of $k_{eff} = 1.362$ represents a substantially, almost overwhelmingly, supercritical physical state. While an untrained observer or layperson might entirely misinterpret this extraordinarily high number as indicative of a severely uncontrollable, inherently dangerous safety hazard, within the operational paradigm of commercial nuclear engineering, this result is not only entirely expected but fundamentally necessary to the economical operation of the baseload facility [7]. The calculated state represents an idealized "cold, unborated, fresh-core" condition, a theoretical maximum completely devoid of the complex, multi-layered standard reactivity control mechanisms actively utilized in the actual Temelín reactor environments.

In practical operational reality, a massive pressurized water reactor such as the VVER-1000 requires a substantial, deliberately engineered reservoir of intrinsic excess reactivity embedded within the fresh fuel matrix. This large reactivity buffer is strictly necessary to compensate for ongoing fuel depletion (physical burnup of U-235), the rapid operational buildup of highly potent neutron-absorbing fission product poisons (most notably the severe xenon-135 and samarium-149 transients), and the vital negative reactivity feedbacks associated with heating the moderator and fuel up to full operating temperatures (the power coefficient of reactivity) [6]. As exhaustively documented in the *NPP Temelín Safety Analysis Reports*, this inherent excess reactivity is meticulously managed and perpetually suppressed through an orchestrated, defense-in-depth chemical and mechanical approach [7][9].

The primary control mechanism relies heavily on the primary coolant system. The highly pressurized water in the VVER-1000 is heavily dosed with soluble boric acid, acting as a potent, homogeneously distributed thermal neutron absorber. According to strict SÚJB regulatory mandates and operational reports provided by ČEZ, the boric acid concentration is finely tuned by the chemical and volume control systems (CVCS) throughout the entire 18-month fuel cycle to maintain a perfect, steady-state k_{eff} of exactly 1.0 at full 100% power, while simultaneously guaranteeing an adequate, legally binding shutdown margin ($k_{eff} < 0.95$) during cold, zero-power shutdown conditions [3][8].

Furthermore, to counteract the massive initial reactivity of the fresh 3.5% to 4.0% enriched fuel batches without relying on dangerously high soluble boron concentrations—which could inadvertently trigger a positive moderator temperature coefficient—modern fuel designs loaded into Temelín utilize integral burnable absorbers. In the advanced Westinghouse RWFA-T and the Russian TVSA-T designs, powerful absorbing isotopes like gadolinium (Gd-157) or dysprosium hafnate are sintered directly into the uranium dioxide pellets of specific fuel rods. These strong absorbers severely depress the localized thermal neutron flux during the critical early stages of the fuel cycle, burning out at a predictable rate that closely matches the depletion of the uranium fuel itself [4][5][10].

6. Conclusions

This comprehensive semester project successfully constructed and validated a deterministic, analytical criticality estimate for the VVER-1000/V-320 pressurized water reactors currently operating at the Temelín Nuclear Power Plant in the Czech Republic. By strictly applying the classical analytical frameworks of the four-factor formula and fundamental one-group neutron diffusion theory, an idealized core configuration with a standard 3.5% average enrichment yielded an infinite multiplication factor (k_{∞}) of 1.384 and a resulting effective multiplication factor (k_{eff}) of 1.362. These mathematically derived, highly supercritical figures authentically reflect the massive excess reactivity intrinsically engineered into commercial nuclear fuel batches necessary to sustain continuous, uninterrupted baseload power production over long operational cycles [5][6].

The final numerical findings inherently underscore the profound physical imperative for rigorous, multi-layered reactivity control measures. The sheer scale of the unborated k_{eff} strongly validates the stringent, non-negotiable regulatory requirements set forth by both SÚJB and European oversight agencies regarding the mandatory deployment of active soluble boron systems, high-speed control rod drive mechanisms, and integral burnable absorbers within the fuel matrix [3][9]. While this simplified analytical model establishes a highly robust, mathematically verifiable foundational intuition regarding the core neutronic behaviors of the specific VVER-1000 system architecture, advanced computational tools—such as 3D stochastic Monte Carlo simulations like MCNP—remain the absolute, indispensable standard for conducting comprehensive, safety-critical nuclear analysis in the modern nuclear industry.

References

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Appendix: Simulation Code (Python)

The following Python script automates the analytical mathematical calculations based on the deterministic point-kinetic methodology explored in this report:

```
import math

def calculate_keff(enrichment_percent, moderator_temp_C):
    # Neutronic constants evaluated at thermal energies (0.0253 eV)
    nu = 2.42
    sigma_f235 = 582.0 # microscopic fission cross-section U-235 (barns)
    sigma_a235 = 680.0 # microscopic absorption cross-section U-235 (barns)
    sigma_a238 = 2.7   # microscopic absorption cross-section U-238 (barns)
    sigma_am = 0.66    # microscopic absorption cross-section H2O (barns)

    # Core structural properties
    V_m_V_f = 1.7      # Moderator to fuel volume ratio
    rho_UO2 = 10.4      # Density of UO2 (g/cm3)
    rho_m = 1.0         # Density of water (g/cm3) at cold state (20 deg C)

    e = enrichment_percent / 100.0
    N_235 = e
    N_238 = 1.0 - e

    # 1. Thermal Fission Factor (eta)
    eta = nu * ((N_235 * sigma_f235) / ((N_235 * sigma_a235) + (N_238 * sigma_a238)))

    # 2. Thermal Utilization Factor (f)
    N_m_N_U = (rho_m / 18.0) / (rho_UO2 / 270.0)
    sigma_a_fuel = (N_235 * sigma_a235) + (N_238 * sigma_a238)
    f = 1.0 / (1.0 + (N_m_N_U * (sigma_am / sigma_a_fuel) * V_m_V_f))

    # 3. Resonance Escape Probability (p) & Fast Fission Factor (epsilon)
    p = 0.75
    epsilon = 1.05

    # Infinite Multiplication Factor
    k_inf = eta * f * p * epsilon

    # 4. Non-leakage probability (P_NL)
    H = 353.0 # Core active height (cm)
    R = 156.0 # Core active radius (cm)
    M2 = 50.0 # Migration area (cm2)
```

```

Bg2 = (2.405 / R)**2 + (math.pi / H)**2
P_NL = 1.0 / (1.0 + M2 * Bg2)

# Effective Multiplication Factor
k_eff = k_inf * P_NL

return {
    "eta": eta,
    "f": f,
    "p": p,
    "epsilon": epsilon,
    "k_inf": k_inf,
    "P_NL": P_NL,
    "k_eff": k_eff
}

if __name__ == "__main__":
    results = calculate_keff(3.5, 20)
    print("Analytical Criticality Estimate for Temelin VVER-1000 (Unborated Cold
Core)")
    print("-" * 75)
    for key, val in results.items():
        print(f"{key:>10} = {val:.5f}")

```